

ECON-7102: Environmental Economics I

Class 2: Introduction to Environmental Economics

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Today's Agenda

1. Syllabus review (questions/concerns?)
2. Introduction to environmental economics
 - Big picture
 - Environmental economics from an intermediate micro perspective
 - Defining economic efficiency
3. For next class...

Overview of course

- Theory of externalities
- Theory of environmental policy
 - Policy instrument choice
 - Pigouvian taxes, command-and-control, tradeable permits
- Welfare analysis
- Non-market valuation
 - Property value models, sorting, health
- Topics
 - Water, utility pricing, electricity externalities and pricing, energy and smart technologies, climate change, conservation, forecasts, domestic climate policy

Fullerton & Stavins, Nature, 1998

- **Myth of the universal market**
 - Market failures are environmental economists' bread and butter
- **Myth of the market solutions**
 - Market solutions can be the most cost-effective solution, but might not always work
- **Myth of market prices**
 - Environmental economists care a lot about goods and services that are not bought/sold in markets
- **Myth of efficiency**
 - Equity and distribution matters too

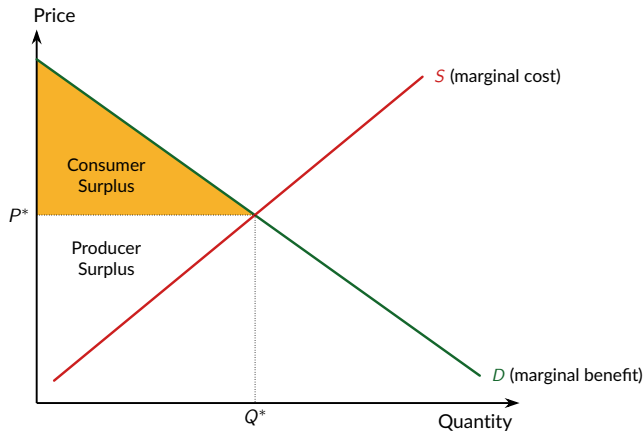
What do environmental economists do?

- Understanding the magnitude of environmental externalities and efficient ways to manage them.
 - Measure benefits of (changes in) environmental quality
 - E.g., how much should BP pay for Deepwater Horizon oil spill?
 - Translate environmental quality changes into dollar terms (willingness to pay for improved air quality)
 - Evaluate alternative environmental policies
 - E.g., design “cap-and-trade” market for reducing air pollution (1990s Acid Rain Program)
 - Generally, env. economists are proponents of “market-based” policies because they achieve *optimal levels of pollution at least cost*
- Characterizing behavior of polluting/regulated firms and individuals affected by environmental policies
- In the last ~20 years, environmental economics has become increasingly empirical.
 - Evident in presentations at 2026 World Congress of Environmental and Resource Economists: [\[Link\]](#)

An intermediate micro introduction to
efficient environmental policy...

Economic efficiency = maximizing net
surplus

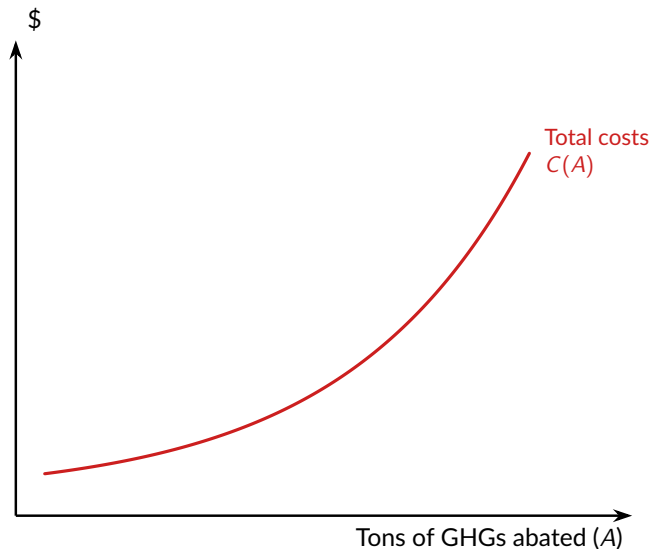
Economic efficiency and net surplus



Refresher on net surplus

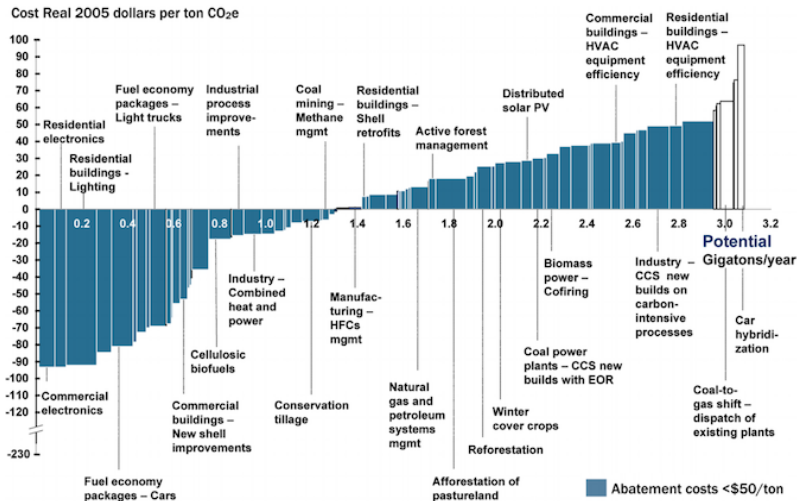
- Equilibrium outcomes (Q^*, P^*) maximize economic surplus $(CS + PS)$.

Costs of reducing GHG emissions



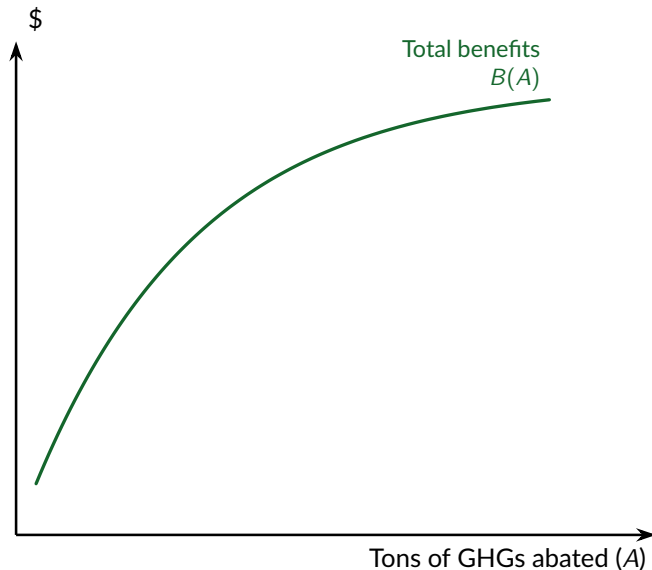
- GHG emissions are a “bad.” They cause climate change.
- GHG abatement (A) is a “good.” This reduces GHG emissions that cause climate change.
- We can model GHG abatement like any other economic good.

Costs of reducing GHG emissions



Source: <https://www.edf.org/revamped-cost-curve-reaching-net-zero-emissions>

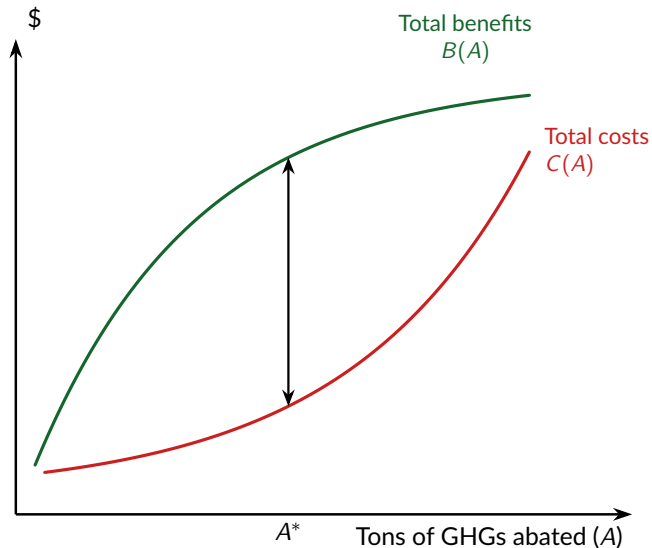
Benefits of reducing GHG emissions



- Abating (or reducing) GHG emissions provides benefits, namely less climate change
- Think of GHG abatement on a range from “business as usual” emissions to “net zero”

What is the optimal amount of pollution?

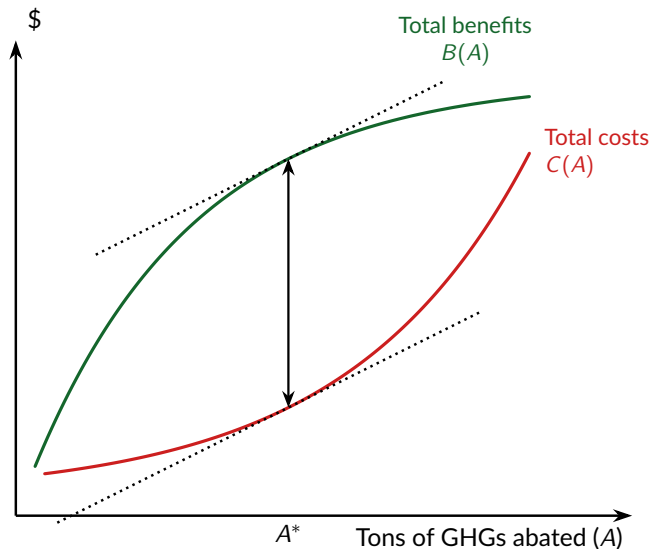
Maximizing net benefits



- We want to adopt a policy that maximize net benefits of reductions of GHGs. That will be an efficient allocation of resources.
- So, find the level of abatement (A^*) where $B(A) - C(A)$ is largest.
- Key insight: the optimal amount of abatement is typically not abating 100% of emissions*
 - Why is this the case?

* Note: for GHGs, we may indeed want to abate all emissions (e.g., "net zero"), or perhaps more than 100%. For most pollutants, the optimal amount of emissions is greater than zero.

Maximizing net benefits

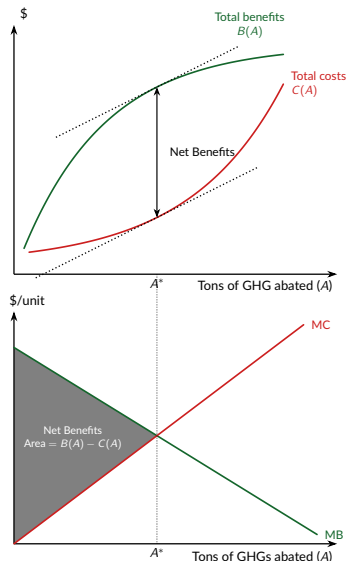


- In “ECON 101” we were taught to set *marginal costs* equal to *marginal benefits*.
- Why? Because that maximizes total net benefits.
- The MC curve is the first derivative of the total cost curve.
- The MB curve is the first derivative of the total benefit curve.

- $\max_A B(A) - C(A)$
- $\rightarrow B'(A^*) = C'(A^*)$

From net benefits to supply and demand

- MB is decreasing because $B(A)$ is increasing at a decreasing rate
 - i.e., diminishing marginal utility
- MC is increasing because $C(A)$ is increasing at an increasing rate
 - i.e., diminishing marginal returns
- Environmental policy/regulation seeks to maximize net benefits.
- Markets do this automatically.



When are markets efficient? (i)

Pareto optimality

An economic outcome is said to be Pareto optimal if a reallocation of resources cannot make at least one person better off without making another person worse off.

(Phaneuf & Requate, 2007, p. 4)

- Focuses on maximizing the size of the pie.
- Does not consider distribution of benefits.

When are markets efficient? (ii)

First Fundamental Theorem of Welfare Economics (an informal version)

If every relevant good is traded in a market at publicly known prices (i.e., if there is a complete set of markets), and if households and firms act perfectly competitively (i.e., as price takers), then the market outcome is Pareto optimal. That is, when markets are complete, any competitive equilibrium is necessarily Pareto optimal.

(Mas-Colell, Whinston, and Green, 1995, p. 308)

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“Any inefficiencies that arise in a market economy, and hence any role for Pareto-improving market intervention, must be traceable to a violation of at least one of the assumptions of this theorem.” (MWG, 1995)

When are markets efficient? (iii)

Second Fundamental Theorem of Welfare Economics (an informal version)

If household preferences and firm production sets are convex, there is a complete set of markets with publicly known prices, and every agent acts as a price taker, then any Pareto optimal outcome can be achieved as a competitive equilibrium if appropriate lump-sum transfers of wealth are arranged.

(Mas-Colell, Whinston, and Green, 1995, p. 308)

- The SWT “gives conditions under which a Pareto optimum allocation can be supported as a price equilibrium with transfers.”
- In other words, lump-sum income transfers can be used for redistributive purposes while supporting an efficient outcome (i.e., reallocating slices of pie without reducing its size).

When are markets efficient? (iv)

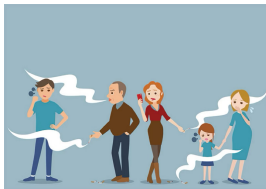
- If markets are complete & perfectly competitive, the FWT suggests that markets will maximize net benefits (e.g., maximize the size of the pie).
 - This is just the classic “invisible hand” argument.
- Similarly, the SWT allows for (potential) redistribution through transfers to satisfy distributional goals.
- Both require complete and perfectly competitive markets.
- Complete markets (*i.e., that all relevant costs and benefits are borne by participants in the market*) requires **no externalities**.

When are markets efficient? (v)

Externalities

An externality exists when agent A's utility or production function depends directly on real variables chosen by another agent B, without an offer of compensation or other attention given to the effect on A's well-being.

(Phaneuf & Requate, 2017)



Concluding thoughts (for now) on efficiency and externalities

- “Complete markets” is violated by the existence of environmental externalities and unpriced “environmental” goods and services.
 - Missing markets can arise because a lack of property rights.
- Efficient outcomes are elusive, but useful benchmarks.
- We will focus (mostly) on environmental externalities, although other market failures are prominent in environmental economics.
 - Imperfect competition
 - Increasing returns to scale (e.g., natural monopolies)
 - Information asymmetry and information failures
 - “Internalities” (i.e., optimization mistakes/errors)
 - Identifying/correcting these market failures are typically the motivation for doing research!

For next class...

Readings for next week are posted on the course page:

caseyjwichman.com/econ7102#week2

- First reading response due for Coase (1960).
- Due: Tuesday, 11:59 PM.
- Submit via Canvas.